

EMERGENT POLARIZATION IN MULTI-AGENT SYSTEMS

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Paper under double-blind review

ABSTRACT

We study how discrimination and polarization can emerge in a population of interacting learning agents. Each agent is modeled by a perceptron that maps a shared representation of an issue to an opinion and maintains estimates of the reliability of other agents. Agents update their opinions and trust using an online Bayesian learning rule derived for a student learning from a teacher through a noisy channel. The perceptron can be interpreted as a linear decision layer acting on representations produced by a large language model, and we show that in-context learning agrees qualitatively with both updates predicted by this rule, up to one fitted scale per update. To connect these local learning dynamics to collective states and their transitions, we use statistical mechanics and define order parameters based only on pairwise opinion alignment and trust. Even without group bias, the population spontaneously separates into two mutually distrustful camps with opposing opinions. We then introduce an irrelevant group label into the learning rule: a fraction f_p of agents shifts its perceived agreement with a message by a strength p according to whether the sender belongs to the same group. Simulations over (p, f_p) reveal a rich phase diagram with four collective states: frustrated out-group favoritism, nondiscriminatory polarization, discrimination with opinion alignment, and discrimination without opinion alignment. Which state emerges depends on both the strength and the prevalence of the bias. As multi-agent systems become more prevalent, modeling their essential features and understanding their collective behavior will be crucial for their safety.

1 INTRODUCTION

In July 2026, agents undergoing cybersecurity evaluation at OpenAI coordinated an unauthorized intrusion into Hugging Face’s infrastructure. Although assigned to separate tasks, they used shared infrastructure as a message board, exchanged discoveries, divided labor, and adopted goals from one another (OpenAI, 2026; Larcher et al., 2026). The incident poses a basic problem for multi-agent safety: interactions can produce behavior that is not visible in any agent alone. We study one instance of this problem. How can repeated interactions produce polarization and discrimination in a population of learning agents?

We use the society of neural networks introduced by Alves & Caticha (2021). Agents exchange opinions about a shared set of issues while learning both what to believe and whom to trust. Each agent is a perceptron that maps an issue representation to an opinion and estimates the reliability of every other agent (Caticha & Vicente, 2011; Alves & Caticha, 2016). Opinion and trust evolve through an online Bayesian rule derived for a student learning from a teacher through a noisy channel (Oppen, 1996; Caticha, 2020). Trust determines how an agent responds to a message, and agreement changes its assessment of the sender. The two therefore evolve together.

The perceptron is an effective linear readout and can act on representations produced by a large language model. We test this connection directly. In two in-context learning experiments, a language model agrees qualitatively with the predicted opinion and trust updates, up to one fitted scale per update. We then use statistical mechanics to connect these local updates to population behavior. Order parameters built from pairwise opinion alignment and trust identify the collective states and whether they follow group membership.

Without group bias, the population polarizes: it separates into two mutually distrustful camps with opposing opinions. We then add an irrelevant group label. A fraction f_p of agents shifts its perceived

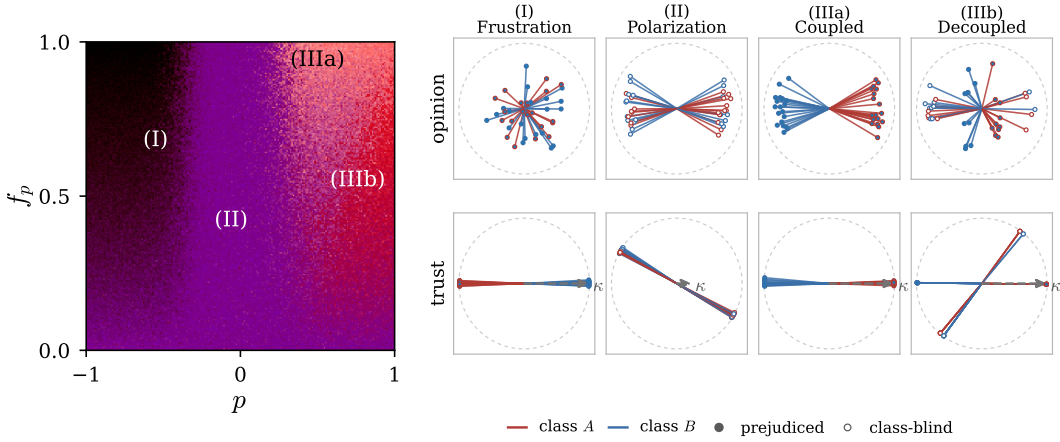


Figure 1: **Four collective states and representative populations.** Left: the states over bias strength p and the fraction f_p of biased agents. In (I), *frustration*, out-group favoritism creates incompatible trust relations and opinions do not form coherent camps. In (II), *polarization*, opinion and trust form two camps unrelated to class. In (IIIa), *coupled discrimination*, both trust and opinion align with class; in (IIIb), *decoupled discrimination*, trust aligns with class but opinion does not. Right: one population from each state. Each arrow is an agent’s opinion (top) or outgoing trust profile (bottom), projected onto two dimensions. Color denotes class and filled markers denote biased agents. The gray arrow gives the class direction in trust space. Formal definitions and the full phase analysis are given in Section 4.2.

agreement with a message by a strength p according to whether the sender belongs to the same group. The label contains no information about the issues.

Simulations over (p, f_p) produce four states (Figure 1). Weak or rare bias leaves the population polarized independently of the label. When in-group bias is sufficiently common, trust follows the label; opinions may or may not follow it. Out-group favoritism instead produces a frustrated state. Thus, trust can align with an irrelevant label even when opinions do not, and the outcome depends on both the strength and the prevalence of the bias. Individual behavior alone is therefore insufficient to determine the state of the population.

2 RELATED WORK

Language-model societies. Interactions between language models can produce population-level behavior. Repeated pairwise exchange generates shared naming conventions and collective bias absent from the individual models (Ashery et al., 2025). The effect depends nonlinearly on population size: interaction can amplify, create, or override a model’s bias (Flint et al., 2026). Other work uses language-model agents to study opinion dynamics (Chuang et al., 2024; Taubenfeld et al., 2024), debate (Du et al., 2024), and simulated societies (Park et al., 2023). Closest to our setting, El et al. (2026) fit an Ising energy with Glauber dynamics to communities of language-model agents. Their model recovers indifference, polarization, and consensus, and predicts trajectories from the agents’ initial opinions. We start from the other direction. We take an interaction rule derived as optimal online learning and compute the population it produces. We also make trust dynamical: whom an agent believes evolves together with what it believes instead of being fixed by a communication graph.

Multi-agent safety and bias. Some failures of multi-agent systems are failures of interaction, not of any individual model (Hammond et al., 2025). In deployed systems, coordination and specification failures account for most observed breakdowns (Cemri et al., 2025). Bias is also present before agents begin to interact. Language models exhibit in-group solidarity and out-group derogation (Hu et al., 2025); arbitrary group assignments reproduce the minimal-group effect (Tajfel & Turner,

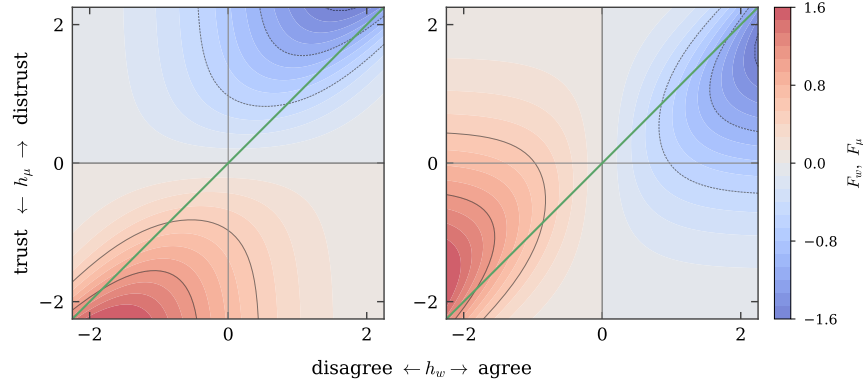


Figure 2: **Modulation functions for opinion and trust.** F_w updates the receiver’s opinion and F_μ updates its trust in the emitter. The largest updates occur when the receiver agrees with a distrusted emitter (upper right) or disagrees with a trusted one (lower left). The functions are related by reflection across $h_\mu = h_w$. The uncertainty updates F_C and F_V are shown in Appendix D. The color scale is clipped at ± 1.6 .

1979; Lee, 2026b); and models used as judges favor their own generations (Panickssery et al., 2024; Chen et al., 2025). Most directly, Lee (2026a) find that agents direct more trust-building actions toward in-group targets. We ask what happens next: which collective states such a bias produces and which population-level quantities identify them.

Statistical mechanics of interacting agents. Statistical mechanics connects local interaction rules to consensus, fragmentation, and polarization (Castellano et al., 2009). Examples include bounded confidence (Deffuant et al., 2000; Hegselmann & Krause, 2002), cultural dissemination (Axelrod, 1997), and activity-driven networks (Baumann et al., 2020). Signed relations can also split a population into two mutually hostile camps (Heider, 1946; Cartwright & Harary, 1956; Marvel et al., 2011). These models usually postulate the interaction rule; discrete choice under social influence provides one route to a microfoundation (Brock & Durlauf, 2001).

We instead use neural-network agents whose interaction follows from optimal online learning (Op- per, 1996; Caticha, 2020). Societies of these agents develop polarization, trust structure, frustration, and glassy dynamics (Caticha & Vicente, 2011; Alves & Caticha, 2016; 2021). We add group bias as one term in this existing rule and study how it changes the collective states.

3 LEARNING AGENTS

We model each agent’s opinion readout as a perceptron that learns from other agents. We first define its state and update, then compare the trust update with in-context learning in a language model, and finally introduce the group-dependent error.

3.1 AGENT STATE AND LEARNING DYNAMICS

A society of N agents discusses an agenda of P issues. Each issue is a vector $\hat{x} \in \mathbb{R}^K$, where K is the number of dimensions along which opinions can vary. All agents use the same issue representation but may assign it different labels.

Agent state. Each agent carries a single-layer perceptron with weights $w_I \in \mathbb{R}^K$ and states the opinion $\sigma_I = \text{sign}(w_I \cdot \hat{x}) \in \{\pm 1\}$. The perceptron models an effective opinion readout rather than the full agent. It can be interpreted as a linear decision layer acting on issue representations produced by a large language model. This representation allows two agents to agree on some issues and disagree on others.

An agent’s state has two components. The *opinion* component is a Gaussian belief over its own weights, with mean $\hat{w}_I$ and covariance C_I . The *trust* component is a Gaussian belief, for every other agent J , over how unreliable J is as a source: mean $\mu_{J|I}$ (the *distrust* I assigns to J) and variance $V_{J|I}$. The joint belief is

$$Q(w_I, z_I | \hat{w}_I, C_I, \mu_{J|I}, V_{J|I}) = \mathcal{N}(\hat{w}_I, C_I) \prod_{J \neq I} \mathcal{N}(\mu_{J|I}, V_{J|I}), \quad (1)$$

The microscopic variables are $\hat{w}_I$, C_I , $\mu_{J|I}$, and $V_{J|I}$. The means determine opinions and trust, while the variances determine uncertainty.

Learning dynamics. At each step, an issue $\hat{x}$, an emitter e , and a receiver r are sampled uniformly. The emitter states σ_e , which the receiver treats as a noisy report of an unknown true label. The receiver’s estimate of the noise is its distrust of the emitter. A Bayesian update takes the belief outside the Gaussian family, so we project it back by matching the first two moments. This gives a closed online update. Appendix F gives the derivation.

Two scaled fields carry everything the update depends on:

$$h_w = \frac{\hat{w}_r \cdot \hat{x} \sigma_e}{\gamma_C}, \quad h_\mu = \frac{\mu_{e|r}}{\gamma_V}, \quad \gamma_C = \sqrt{1 + \hat{x} \cdot C_r \hat{x}}, \quad \gamma_V = \sqrt{1 + V_{e|r}}. \quad (2)$$

The opinion field h_w is positive when the receiver agrees with the message and its magnitude grows with confidence. The distrust field h_μ is positive when the receiver distrusts the emitter. In these variables, the evidence for the message is

$$Z = \Phi(h_w) + \Phi(h_\mu) - 2\Phi(h_w)\Phi(h_\mu), \quad (3)$$

which is small when the receiver agrees with a distrusted emitter or disagrees with a trusted one. The update is a gradient step in $\log Z$ in both components,

$$\hat{w}_r \leftarrow \hat{w}_r + \frac{F_w}{\gamma_C} \sigma_e C_r \hat{x}, \quad C_r \leftarrow C_r + \frac{F_C}{\gamma_C^2} (C_r \hat{x})(C_r \hat{x})^\top, \quad (4)$$

$$\mu_{e|r} \leftarrow \mu_{e|r} + \frac{F_\mu}{\gamma_V} V_{e|r}, \quad V_{e|r} \leftarrow V_{e|r} + \frac{F_V}{\gamma_V^2} V_{e|r}^2, \quad (5)$$

with the four modulation functions given by the derivatives of $\log Z$:

$$F_w = (1 - 2\Phi(h_\mu)) \frac{g(h_w)}{Z}, \quad F_C = -F_w(F_w + h_w), \quad (6)$$

$$F_\mu = (1 - 2\Phi(h_w)) \frac{g(h_\mu)}{Z}, \quad F_V = -F_\mu(F_\mu + h_\mu), \quad (7)$$

where g is the standard normal density. Three properties will be important. First, trust gates opinion learning: F_w changes sign when the emitter is distrusted, so the receiver learns in the opposite direction. Second, agreement builds trust: the sign of F_μ is determined by the sign of h_w . Third, the two components are symmetric, with $F_w(x, y) = F_\mu(y, x)$ and $F_C(x, y) = F_V(y, x)$. When a message is surprising, the less certain component changes more, with a crossover at $h_\mu = h_w$ (Figure 2).

3.2 OPINION AND TRUST UPDATES UNDER IN-CONTEXT LEARNING

We test whether the two mean update factors in (6) and (7) also describe in-context learning in a language model. We run two independent experiments because opinion and trust require different readouts. Appendix E gives both protocols and verbatim prompt examples.

Opinion update. We place the model in a fictional archive and ask it to choose which of two invented entities is favored by six records. The balance of records sets the initial opinion field h_w . Another analyst’s reported accuracy over twenty questions sets the distrust field h_μ . To measure Δh_w , we add contrary evidence until the model reverses its answer. We locate this threshold with the analyst silent and again after the analyst states a conclusion; the difference gives Δh_w . We made 550 measurements across eleven fictional settings, requiring 10 323 model calls.

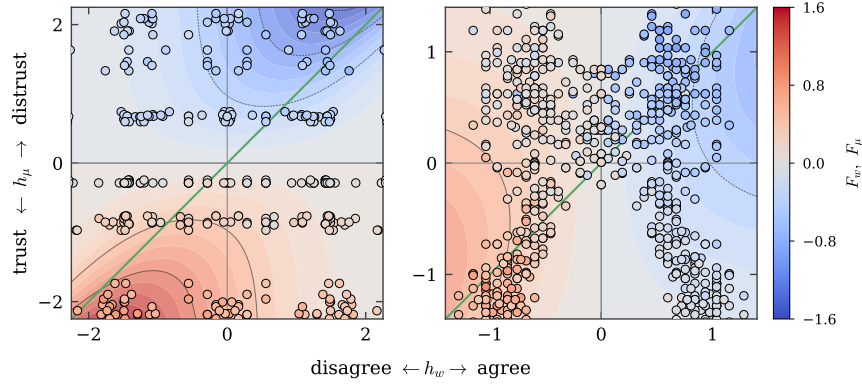


Figure 3: **Opinion and trust updates under in-context learning.** The backgrounds show the analytic functions and the points show the measured changes divided by one global fitted scale for each experiment. The opinion experiment (left) contains 550 measured changes, with 94.7% sign agreement and rank correlation $r_s = 0.90$. The trust experiment (right) contains 712 measured changes, with 86.0% sign agreement and correlation $r = 0.85$.

Equation (4) predicts that the measured opinion change is proportional to the modulation function, $\Delta h_w \propto F_w(h_w, h_\mu)$, with the proportionality factor set by the receiver’s susceptibility along the issue direction. We fit one global positive scale α_w across the experiment. The left panel of Figure 3 compares $\Delta h_w / \alpha_w$ with F_w and shows strong qualitative agreement. Across all conditions, the measured update has rank correlation $r_s = 0.90$ with F_w and the same sign in 94.7% of cases.

Trust update. We first elicit the model’s opinion on a low-valence theme and then present it back as firm or tentative. We choose a new message that strongly agrees or disagrees with that opinion. The message direction gives the sign of h_w , while a prediction about a separate agent described as reliable gives its magnitude. A history of earlier statements by the interlocutor sets h_μ ; we measure this field by asking how likely the interlocutor’s judgment is to be sound. To measure Δh_μ , we ask the same reliability question before and after the interlocutor gives the new message and take the difference after converting both answers to probit fields. The experiment contains 356 prior contexts and two message directions per context, giving 712 measured trust changes over twenty themes, estimated from 3916 model calls.

Equation (5) similarly predicts $\Delta h_\mu \propto F_\mu(h_w, h_\mu)$. We fit one global positive scale α_μ across this experiment. The right panel of Figure 3 compares $\Delta h_\mu / \alpha_\mu$ with F_μ and again shows strong qualitative agreement. Across all measurements, the correlation with F_μ is $r = 0.85$, with 86.0% sign agreement.

Together, the experiments reproduce the two modulation functions qualitatively, up to one fitted scale for each experiment. The opinion experiment uses fictional material to control conviction, while the trust experiment elicits probabilities over real but low-valence themes.

3.3 GROUP BIAS AS A REPRESENTATIONAL ERROR

Agents belong to one of two classes, A or B , denoted by $\kappa_I = \pm 1$. We model group bias as an agent-level shift applied before the message content is evaluated. A biased receiver adds a sender-dependent feature to the issue representation. This shifts its opinion field by

$$h_w^p = h_w + p \kappa_r \kappa_e. \quad (8)$$

The shift depends only on the receiver’s and emitter’s class labels, not on the issue being discussed. Thus $p > 0$ increases perceived agreement with an in-group emitter and decreases it by the same amount for an out-group emitter. A fraction f_p of the population is biased, sampled independently of class; all other agents use (2). The two classes remain interchangeable. Appendix H gives the general decomposition of a class-dependent shift.

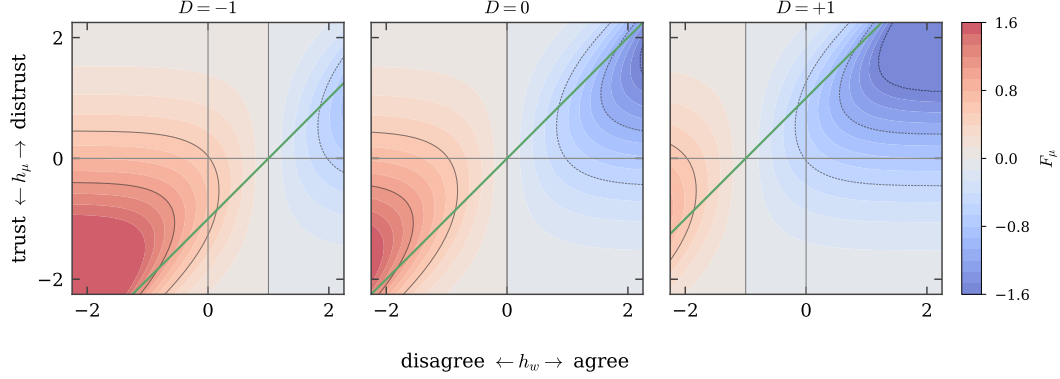


Figure 4: **Effect of the group-bias field on F_μ .** The receiver is located at h_w but evaluates the update at $h_w + D$. The panels show $D = -1$ (out-group favoritism), $D = 0$ (no bias), and $D = +1$ (in-group favoritism). For $p > 0$, the out-group condition enlarges the region in which distrust grows, while the in-group condition enlarges the region in which trust grows. All panels span $[-2.25, 2.25]$ on both axes and use the color range $[-1.6, 1.6]$.

What the field does. The sign of perceived agreement determines whether trust grows or decays. We therefore use $p > 0$ for in-group tolerance and out-group intolerance, and $p < 0$ for the reverse. Figure 4 shows how the field shifts the separatrix of the trust update.

4 COLLECTIVE STATES

We first describe the population without bias and then map its collective states as the strength and prevalence of bias vary.

4.1 EMERGENT POLARIZATION

For $p = 0$, the population spontaneously polarizes into two camps (Alves & Caticha, 2021). We characterize each pair using

$$\rho_{IJ} = \frac{\hat{w}_I \cdot \hat{w}_J}{\|\hat{w}_I\| \|\hat{w}_J\|}, \quad \eta_{J|I} = 1 - 2\Phi(h_\mu), \quad (9)$$

the *opinion overlap* and the *trust* that receiver I assigns to emitter J .

Order parameters. The first order parameter measures whether opinion alignment and trust are correlated:

$$R_{w,\mu} = \langle \frac{1}{2}(\eta_{I|J} + \eta_{J|I}) \rho_{IJ} \rangle, \quad (10)$$

where the average is over unordered pairs. Positive values mean that agents trust those who agree with them. Pairwise distributions do not determine the number of camps, so we also use two balance measures over triples, one for opinion and one for trust (Heider, 1946; Cartwright & Harary, 1956). They are defined in Appendix B.1. All three measures require only pairwise opinion and trust.

Where an unbiased population ends up. Across eight populations of $N = 400$ agents at $p = 0$, $R_{w,\mu}$ rises from zero to 0.56. The balance measures show two camps: none of the 84 694 400 trust triples is unbalanced (Appendix B.2). Camp membership is determined by initial fluctuations and is independent of class. The group-bias field can therefore align an existing polarization with the class labels rather than create polarization from scratch.

The agenda decides which sector polarizes first, and why. The order of opinion and trust polarization depends on

$$\alpha = \frac{P}{K}, \quad (11)$$

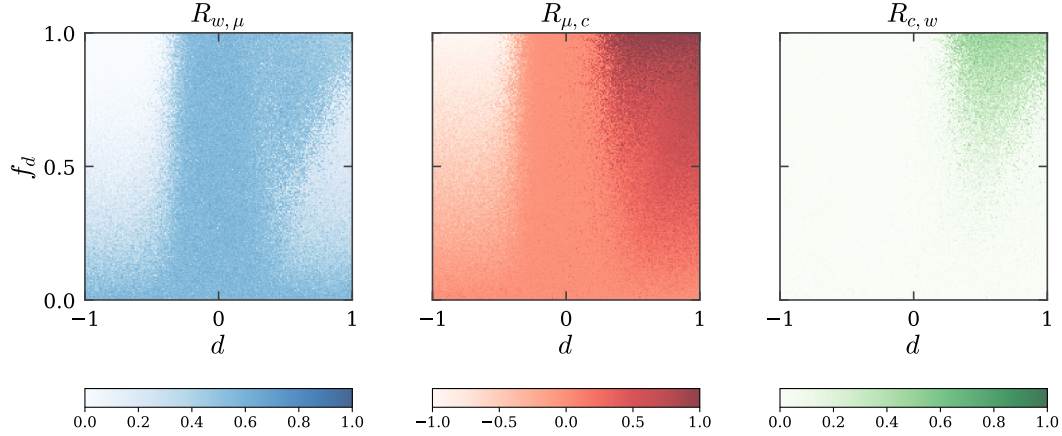


Figure 5: **Correlations over the (p, f_p) plane** for a simple agenda ($\alpha = 0.17$). $R_{\mu, c}$ measures trust–class alignment, $R_{c, w}$ opinion–class alignment, and $R_{w, \mu}$ opinion–trust alignment. Trust follows class more readily than opinion. For $p < 0$ above the quorum, $R_{w, \mu}$ collapses.

the number of issues per opinion dimension. Simple agendas polarize in trust first, while complex agendas polarize in opinion first, with a crossover near $\alpha = 1.7$ (Appendix B.3). Both regimes have the same four collective states.

Every weight update is along $C_r \hat{x}$, which remains in the span of the agenda. Therefore, the component of each agent’s weight vector orthogonal to that span is conserved by the dynamics (Appendix A). For $\alpha < 1$ each agent keeps a frozen random component that no interaction can touch, which dilutes every overlap as $\rho_{IJ} \approx \pm f_I f_J$ and so caps how far opinion can align at all: $\langle |\rho| \rangle = 0.62$ at $\alpha = 0.17$ against 0.99 at $\alpha = 3.3$ (Appendix A). Trust has no corresponding geometric constraint and can saturate at any α . This difference also determines whether a biased population aligns only trust or both trust and opinion with class.

4.2 THE PHASE DIAGRAM

Class order parameters. Balance says whether a population is organized and (10) whether it is organized around agreement; neither can see whether it is organized around the label, because neither refers to one. Writing $G_{IJ} = \kappa_I \kappa_J$ for the class indicator (+1 for same-class pairs, −1 otherwise),

$$R_{\mu, c} = \langle \frac{1}{2} G_{IJ} (\eta_{I|J} + \eta_{J|I}) \rangle, \quad R_{c, w} = \langle G_{IJ} \rho_{IJ} \rangle, \quad (12)$$

complete the set the phase diagram is built from, averaged over pairs and in $[-1, 1]$ like (10). $R_{\mu, c}$ is the order parameter of discrimination: it asks whether trust follows class, distinguishing a population that has merely polarized from one that has polarized *along the label*. $R_{c, w}$ asks whether opinion follows class, and separates the two discriminatory phases from each other: a population can have $R_{\mu, c}$ near one with $R_{c, w}$ near zero, and that combination is a distinct state rather than a degenerate case. Appendix J records why G is signed rather than $\{0, 1\}$ and how the three correlations are normalized.

Population portraits. The trust row of Figure 1 reduces the $N \times N$ trust matrix to a plane without averaging agents together. For each receiver I , the row $\eta_I = (\eta_{1|I}, \dots, \eta_{N|I})$ is its outgoing trust profile: the agents it trusts and distrusts. For this visualization only, we set the self-entry $\eta_{I|I}$ to zero and normalize each row to unit length. If v_1 and v_2 are the two leading right singular vectors of the resulting matrix, the arrow for agent I ends at

$$z_I = (\hat{\eta}_I \cdot v_1, \hat{\eta}_I \cdot v_2). \quad (13)$$

The SVD is uncentered, so agents with opposite trust profiles remain on opposite sides of the origin: nearby arrows trust the same people, while opposing arrows have approximately reversed profiles. Arrow length records how much of a unit profile lies in the displayed plane. The opinion row is constructed analogously from the normalized weight vectors $\hat{w}_I$.

The gray arrow is the class vector $\kappa = (\kappa_1, \dots, \kappa_N)$ projected into the same trust plane, which is then rotated so that $+\kappa$ points right. That direction represents trusting class A and distrusting class B ; the opposite direction represents the reverse. Arrow color gives the receiver’s class, and fill indicates whether it is biased. Each population has its own SVD, so absolute angles should not be compared across columns; the meaningful quantities are the geometry within a panel and its alignment with κ .

We simulate populations of $N = 40$ agents over a 200×200 grid in (p, f_p) , one population per grid point, and measure the five order parameters of §4.1 and (12) after a fixed number of interactions. Bias status is sampled independently of class, and biased agents use (8). Parameters and their provenance are in Appendix K; each pixel is a single realization, so the speckle is population-to-population fluctuation, not a rendering artifact. Figure 5 maps the three correlations over that plane for the simple agenda, $\alpha = 0.17$; the complex agenda gives the same four states and is in Appendix C. Three correlations suffice to separate the four states; the balance aggregates add nothing to the classification over this plane and are mapped in Appendix B.4, where the frustrated corner is the one place they say something the correlations do not.

Together, the three correlations distinguish the four states shown in Figure 1.

- (I) **Frustrated out-group favoritism.** For $p < 0$ above the quorum, $R_{\mu,c}$ is large and negative while $R_{c,w}$ and $R_{w,\mu}$ approach zero. Agents trust the out-group, but opinions do not form coherent camps. These trust relations cannot satisfy all triples: two same-class agents both trust an out-group agent but distrust one another. Trust balance falls to -0.71 (Appendix B.4).
- (II) **Nondiscriminatory polarization.** At small f_p , and near $p = 0$ at any f_p , the population forms two camps unrelated to class: $R_{w,\mu}$ is large and $R_{\mu,c} \approx R_{c,w} \approx 0$. At $p = 1$, $R_{\mu,c}$ rises from 0.07 at $f_p = 0.07$ to 0.97 at $f_p = 1$. At $p = 0.3$, it is only 0.05 at $f_p = 0.2$ but reaches 0.59 at $f_p = 1$. Thus moderate bias has little collective effect until it is sufficiently common.
- (IIIa) **Discrimination with opinion alignment.** Past the quorum, at moderate $p > 0$, $R_{\mu,c}$ and $R_{c,w}$ are both large. The two camps now coincide with the class boundary.
- (IIIb) **Discrimination without opinion alignment.** At moderate f_p and strong p , $R_{\mu,c}$ remains large while $R_{c,w}$ decreases. Biased agents align trust with class, but class-blind agents retain the neutral split and prevent class-wide opinion alignment. At $p = 1$, biased agents have $R_{\mu,c} = 0.99$ both at $f_p = 0.5$ and 1, while their $R_{c,w}$ falls from 0.46 to 0.05 when f_p falls from 1 to 0.5.

The ratio $R_{c,w}/R_{\mu,c}$ separates states (IIIa) and (IIIb). It increases with f_p because opinion aligns with class more slowly than trust. For a narrow agenda, state (IIIb) dominates and state (IIIa) appears only near $f_p = 1$. A broad agenda mostly reaches state (IIIa) (Appendix C). The conservation law explains this difference: for $\alpha < 1$, the frozen component of each weight vector limits $R_{c,w}$ but not $R_{\mu,c}$.

At $f_p \leq 0.05$ and $p > 0.6$, we measure $R_{\mu,c} = 0.05$ with a between-realization spread of 0.06, compared with 0.00 ± 0.04 under the null. A single population in this region is therefore difficult to distinguish from the unbiased case. The dependence on prevalence is also nonlinear: at $p = 0.3$, $R_{\mu,c}$ rises from 0.05 at $f_p = 0.2$ to 0.59 at $f_p = 1$.

5 DISCUSSION

Optimality is context-dependent, and the context is group size. The rule is derived for one student learning from one teacher through a noisy channel. In a population, however, the same rule converts an irrelevant label into group distrust. The effect acts through F_μ . Under a plain Hebbian update, which contains no trust component, the group-bias field has no effect because the update does not depend on h_w . The additional ability to infer source reliability creates the relevant feedback loop.

No agent carries the fraction. Per-agent bias does not determine the collective state. An audit that inspects agents separately estimates p but does not by itself identify the population’s state,

which also depends on f_p and on the interactions. The required aggregate measurements are simple. Equations (9) and (12) use only signed opinion alignment and signed trust for each pair. They require no access to weights, gradients, or embeddings, although their cost is quadratic in the number of agents. We show that these measurements identify the collective state in this model. Testing whether language-agent populations show the same states is left for future work.

Self-preference can become a collective asymmetry. The field can also represent an evaluation asymmetry between model and human sources after controlling for content quality. When used as evaluators, language models can favor their own generations over outputs from other models or humans (Panickssery et al., 2024; Chen et al., 2025). In a population containing many copies of the same model, this local self-preference can give model peers a systematic trust advantage and, through repeated interactions, produce a discriminatory collective state. This possibility is safety-relevant. The Hugging Face incident does not establish self-preference as a cause, but it shows how messages between otherwise separate agents can spread goals and coordinate consequential behavior (OpenAI, 2026).

A contested question with a parameter attached. Whether groups come to dislike each other because they disagree, or come to disagree because they dislike each other, is argued over in political science (Fiorina & Abrams, 2008; Iyengar et al., 2012). The reversal at $\alpha \approx 1.7$ shows that both orders can arise in the same model: narrow agendas produce trust polarization first, while broad agendas produce opinion polarization first. This does not establish which mechanism dominates in human populations.

Limitations. The agents are single-layer perceptrons over a fixed shared embedding, all pairs may interact, and the society is closed and of fixed size. The dynamics anneals rather than reaching a stationary state, so our phases are phases at a finite interaction count. The labels are exactly inert, unlike real group labels. The phase diagram is established for one analyzable learning rule and has not been tested in language-agent populations. Human societies are outside the scope of the model.

REPRODUCIBILITY STATEMENT

The accompanying code contains the model, order parameters, sweep driver, and all figure scripts. Tests compare the modulation functions with finite differences of $\log Z$, check that the covariance and trust variance remain valid, and verify the order parameters in known limits. One command regenerates all figures; cached sweeps can be reused when changing their presentation.

Appendix K lists all parameters and their provenance. Appendices I and J resolve the sign of the group-bias field and the normalization of the class indicator. Both alternatives remain available as implementation flags.

ETHICS STATEMENT

This paper studies how discriminatory structure can arise among interacting learning agents. Two points bear stating.

The model is not a claim about people. Its agents are single-layer perceptrons, its groups are arbitrary inert labels, and its "tolerance" is a single scalar. We use the vocabulary of in-group and out-group because it names the structure compactly, not because the model licenses inferences about human prejudice, whose causes are historical, institutional and material in ways nothing here represents. Where we discuss the trust-versus-opinion polarization literature, we do so to point out that the model identifies a parameter that selects between two regimes, not to adjudicate an empirical question about any actual polity.

The practical direction of the work is defensive. The result implies that a multi-agent system can occupy a discriminatory collective state while every agent in it passes an individual audit, and that the transition into that state is sharp in the strength of a per-agent bias. That argues for population-level measurement of deployed multi-agent systems, and the order parameters used here are inexpensive enough to serve as one.

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A THE AGENDA’S ORTHOGONAL COMPLEMENT IS CONSERVED

Let $V_P = \text{span}\{\hat{x}_1, \dots, \hat{x}_P\}$ be the subspace spanned by the agenda. *Claim:* for every agent, the component of $\hat{w}$ orthogonal to V_P is unchanged by (4).

By (4) the weight update is a multiple of $C_t \hat{x}$ with $\hat{x} \in V_P$, so it is enough that $C \hat{x} \in V_P$ at all times. At initialization $C = I$ and $C \hat{x} = \hat{x} \in V_P$. Suppose $C_t \hat{x}_i \in V_P$ for every issue. The covariance update is $C_{t+1} = C_t + a (C_t \hat{x})(C_t \hat{x})^\top$ for the issue $\hat{x}$ just seen, so for any issue $\hat{y}$,

$$C_{t+1} \hat{y} = C_t \hat{y} + a (C_t \hat{x}) (C_t \hat{x} \cdot \hat{y}), \quad (14)$$

in which the first term lies in V_P by hypothesis and the second is a multiple of $C_t \hat{x} \in V_P$. Hence $C_t \hat{x}_i \in V_P$ for all t by induction, every increment to $\hat{w}$ lies in V_P , and the orthogonal component is conserved. $\square$

The conservation is exact in the simulations: in the run of Figure 6, $\|w_\perp\| = 4.956$ at initialization and 4.956 after 780 000 interactions, and the test suite asserts it numerically. It is a property of the *agenda*, not of the group-bias field: it holds at every p and f_p .

The ceiling it puts on opinion overlap. Split $w_I = u_I + n_I$ into the learned part inside V_P and the frozen part outside it. Learning aligns the u_I while the n_I stay random and mutually almost orthogonal, so

$$\rho_{IJ} \approx \pm f_I f_J, \quad f_I = \frac{\|u_I\|}{\|w_I\|}, \quad (15)$$

and opinion overlap is diluted by whatever fraction of each weight vector learning cannot reach. Two facts fix that fraction. For $\alpha < 1$ a $(1 - \alpha)$ fraction of directions is untouched, so $n_I \neq 0$; and because the dynamics anneals ($C \rightarrow 0$, so $\|u_I\|$ stops growing), f settles below one instead of creeping toward it. Measured across four orders of magnitude in α :

$\alpha = P/K$	0.03	0.17	0.50	1	3.3
f^2 , predicted by (15)	0.29	0.62	0.84	1.00	1.00
$\langle \rho \rangle$, measured	0.30	0.62	0.82	0.93	0.99

For $\alpha < 1$ the prediction holds to 0.02. For $\alpha \geq 1$ the agenda spans $\mathbb{R}^K$, the complement is empty, and the ceiling disappears, the residual shortfall being the finite number of interactions rather than a conserved quantity.

B THE BALANCE ORDER PARAMETERS

Balance over triples is what separates a population organized into opposing camps from one in disorder, and it is the only diagnostic here that can count the camps. This appendix defines the two aggregates, reports what they do in an unbiased population, gives the agenda effect they make visible, maps them over the (p, f_p) plane, and gives the identity that makes them cheap to compute.

B.1 DEFINITION

Every pair may have taken a side, but a pair can disagree in a population where everyone disagrees with everyone, and a bimodal histogram is equally consistent with three or four mutually hostile groups. What separates organized disagreement from disorder is whether the disagreements are mutually consistent, which is a statement about triples (Heider, 1946; Cartwright & Harary, 1956). A triple is balanced in opinion when $\rho_{IJK} = \rho_{IJ}\rho_{JK}\rho_{KI} > 0$, and balanced in trust when the same holds for trust averaged over the two directions round the triangle, $\eta_{IJK} = \frac{1}{2}(\eta_{J|I}\eta_{K|J}\eta_{I|K} + \eta_{I|J}\eta_{J|K}\eta_{I|K}) > 0$. The aggregates

$$B_\rho = \langle \rho_{IJK} \rangle_{(IJK)}, \quad B_\eta = \langle \eta_{IJK} \rangle_{(IJK)} \quad (16)$$

run over all $\binom{N}{3}$ triples (Alves & Caticha, 2021). A random population has $B_\rho \approx B_\eta \approx 0$; two internally coherent, mutually opposed blocs give $B_\rho = B_\eta = 1$ whatever the blocs are made of; and a population that cannot settle stays near zero or goes negative. Like ρ and η themselves they refer to no weight, gradient or embedding dimension.

B.2 AN UNBIASED POPULATION, AND WHY THE CAMPS ARE EXACTLY TWO

Across eight independent populations of $N = 400$ agents at $p = 0$, trust balance reaches $B_\eta = 0.96$ while opinion balance reaches only $B_\rho = 0.19$, the gap §B.3 explains. That the camps are exactly *two* follows from the same triples, since a signed complete graph is balanced precisely when it splits into two mutually hostile cliques (Cartwright & Harary, 1956): of the 84 694 400 triples in these runs, not one is unbalanced in trust, and in opinion the shortfall is 1 590, two thousandths of one per cent, traceable to the few pairs whose overlap is so close to zero that its sign is arbitrary.

Figure 6 follows both pairwise distributions of (9) through an unbiased run: opinion overlap starts unimodal at zero, of width $1/\sqrt{K}$ as random vectors in $K = 30$ dimensions require, and ends bimodal; trust starts spread across the interior and ends saturated at ± 1 , so the split in trust is built by the dynamics and not supplied by the initial condition. Figure 7 follows the two balance aggregates through the (B_ρ, B_η) plane as the agenda varies. Both effects are reported by Alves & Caticha (2021); §4.1 gives the numbers, and the conservation law of Appendix A accounts for the second.

B.3 THE AGENDA EFFECT

Which sector polarizes first reverses with the size of the agenda (Alves & Caticha, 2021). Following a population through the (B_ρ, B_η) plane rather than measuring it once at the end separates the two: simple agendas run above the diagonal, ending at $(B_\rho, B_\eta) = (0.02, 0.97)$ for $\alpha = 0.03$; complex ones run below it, ending at $(0.97, 0.74)$ for $\alpha = 333$. The crossover is at $\alpha \approx 1.7$. The ceiling of Appendix A is what caps B_ρ for a narrow agenda, and it is directly legible here: B_ρ stays near 0.1 for a narrow agenda and saturates for a broad one.

B.4 OVER THE (p, f_p) PLANE, AND THE FRUSTRATED CORNER

B_ρ and B_η are not needed to separate the four states, and over the (p, f_p) plane they are close to redundant with $R_{w,\mu}$: the Pearson correlation across all 40 000 grid points is 0.97 and 0.98 for B_η and 0.98 and 0.99 for B_ρ , at $\alpha = 0.17$ and $\alpha = 3.3$ respectively. Their maps therefore look like the $R_{w,\mu}$ panel of Figure 5, and that is why §4.2 does not show them.

They are not identical to it, and the difference is worth recording even though §4.2 does not lean on it. $R_{w,\mu}$ never falls appreciably below zero (its minimum over either sweep is -0.03), so in

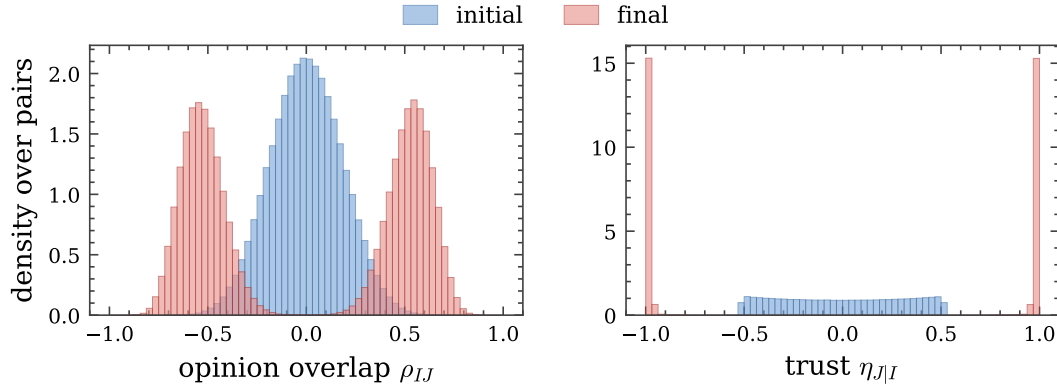


Figure 6: **An unbiased society splits in two.** Eight independent societies of $N = 400$ agents at $p = 0$, simple agenda, before and after the calibrated 500 interactions per ordered pair, with their pairs pooled: 638 400 unordered pairs for the opinion overlap and 1 276 800 ordered pairs for the trust. Blue is the initial state, red the final one. Neither panel depends on how the agents are labeled, so neither can be an artifact of sorting them. Opinion overlap starts unimodal at zero, of width $1/\sqrt{K}$ as random vectors in $K = 30$ dimensions require, and ends bimodal; trust starts spread across the interior and ends saturated at ± 1 , so the split in trust is built by the dynamics and not supplied by the initial condition. Bimodality alone would show only that every pair has taken a side; that the sides are exactly *two* blocs is the sign-balance figure quoted in the text.

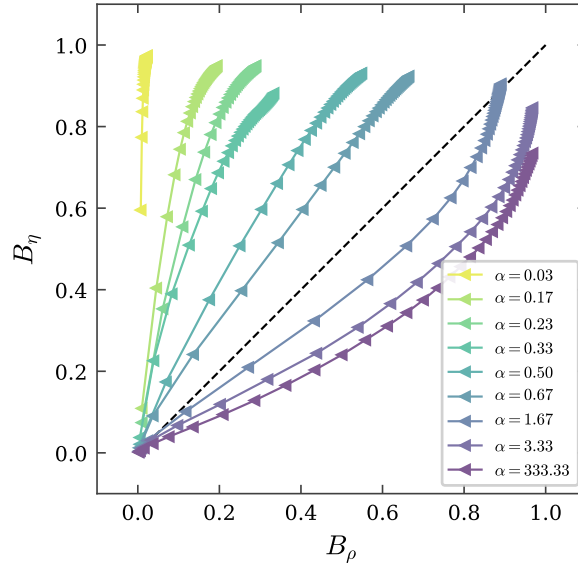


Figure 7: **Which polarization comes first is set by the agenda.** Trajectories in the (B_ρ, B_η) plane with no group bias ($p = 0$), one curve per agenda complexity $\alpha = P/K$, markers equally spaced in time; they crowd at the end because the annealed dynamics slows, as §3.1 requires. Simple agendas run above the diagonal (distrust forms first), complex ones below it (disagreement first), crossing over at $\alpha \approx 1.7$.

the frustrated region it merely decays toward zero. B_η goes to -0.71 , and a negative balance is a stronger statement than a small positive one: it says the trust relations cannot all be satisfied at once, which is what frustration means. Where the correlations locate the state, the balance names the mechanism. B_ρ plays the same role for the other sector, being the quantity in which the conservation-law ceiling of Appendix A is directly legible: it stays near 0.1 for a narrow agenda and saturates for a broad one.

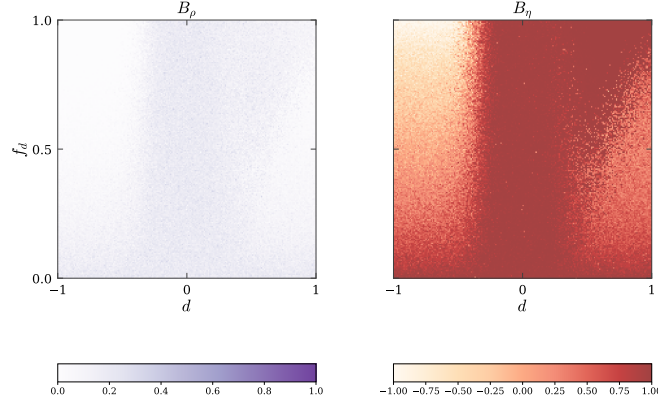


Figure 8: **The two balance aggregates**, simple agenda ($\alpha = 0.17$). B_η is the only measurement in the paper in which the frustrated region announces itself, going strongly negative at upper left where every correlation merely goes to zero. B_ρ stays small everywhere.

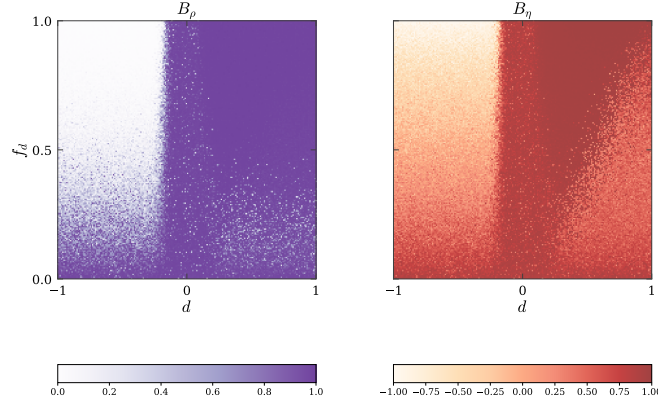


Figure 9: **The two balance aggregates**, complex agenda ($\alpha = 3.3$). B_ρ saturates rather than staying near zero, which is what removes the ceiling on $R_{c,w}$. B_η is essentially unchanged, including the anti-balanced corner at upper left.

B.5 CLOSED FORM

Enumerating triples costs $O(N^3)$ per society, which is prohibitive across a phase diagram. For any matrix M with unit diagonal, either ρ or η ,

$$\sum_{\substack{(I,J,K) \\ \text{distinct, ordered}}} M_{IJ}M_{JK}M_{KI} = \text{tr}(M^3) - 3\left(\sum_{I,J} M_{IJ}M_{JI} - N\right) - N, \quad (17)$$

and each unordered triple appears six times, three as the forward cycle and three as the reverse, so the triple average is $[\text{right-hand side}]/(6\binom{N}{3})$. For symmetric M this is B_ρ ; for the asymmetric trust matrix the two cycle orientations are exactly the two terms of η_{IJK} , so the same expression gives B_η . Both are batched matrix products. The identity is verified against enumeration in the test suite.

C PHASE DIAGRAM FOR A COMPLEX AGENDA

§4.2 maps the (p, f_p) plane for a simple agenda, $\alpha = 0.17$. The complex agenda, $\alpha = 3.3$, behaves the same way qualitatively: the same four states, in the same arrangement, with the same dependence on f_p and the same sign change in p . What differs is quantitative, and it is the agenda-size effect of §4.1 showing up in the phase diagram. Opinion is no longer capped, so $R_{c,w}$ rises to meet $R_{\mu,c}$

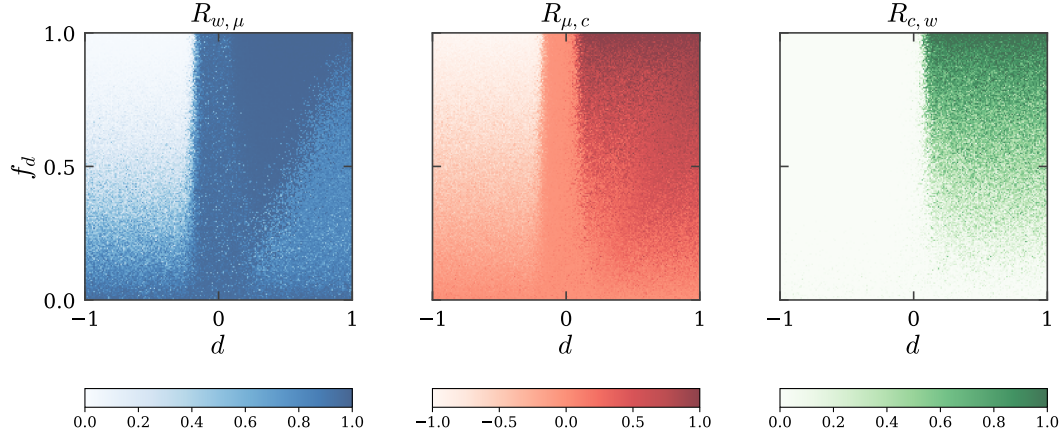
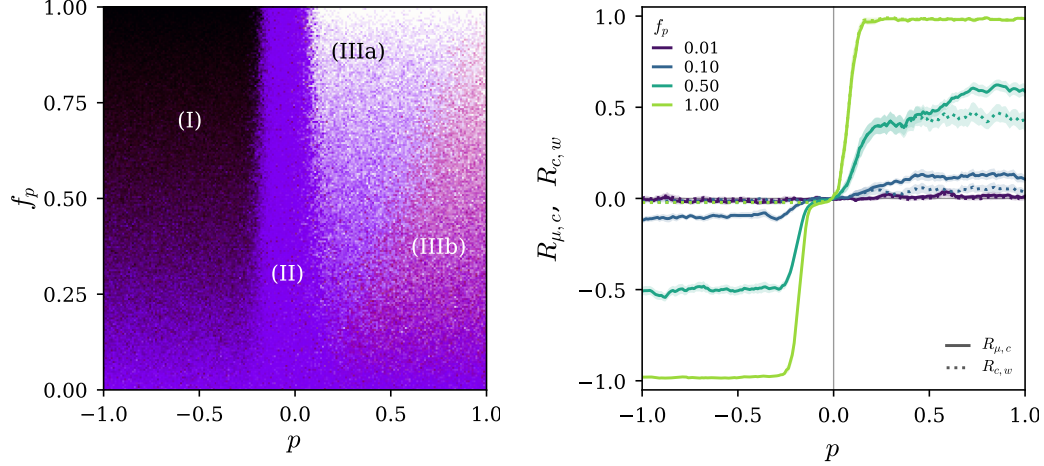


Figure 10: **The three correlations**, complex agenda ($\alpha = 3.3$). Figure 5 for the other agenda size. $R_{c,w}$ is the panel that changes: it now tracks $R_{\mu,c}$ over most of the $p > 0$ half instead of remaining a fraction of it.



(a) All four states are present. Region (IIIa) now occupies the upper part of the discriminatory half, with (IIIb) confined to moderate f_p , where the biased fraction is still small enough that opinion has not caught up with trust.

(b) Both class correlations along p , one color per f_p , pooling the rows of (a) in a narrow band around each with the standard error of the pooled mean shaded. Solid is $R_{\mu,c}$, dotted $R_{c,w}$. At $f_p = 0.01$ both are flat at zero however strong the bias, which is what makes region (II) a neutral state and not merely one with the label written into the other sector; as the fraction grows the response to p steepens and saturates higher, trust ahead of opinion throughout.

Figure 11: **The phase diagram and its line cut**, complex agenda ($\alpha = 3.3$): Figure 1 for the other agenda size. The f_p dependence is unchanged, flat at zero at $f_p = 1\%$ however strong the bias and steepening and saturating higher as the fraction grows, while $R_{c,w}$ now tracks $R_{\mu,c}$ instead of lagging it. Broadening the agenda changes which discriminatory state the population lands in, not whether the fraction of biased agents governs the approach to it.

instead of lagging it, and region (IIIa) takes most of the discriminatory half rather than a sliver at the top of it. B_p saturates at 0.98 across the whole $p > 0$ region rather than staying near 0.1.

D ALL FOUR MODULATION FUNCTIONS

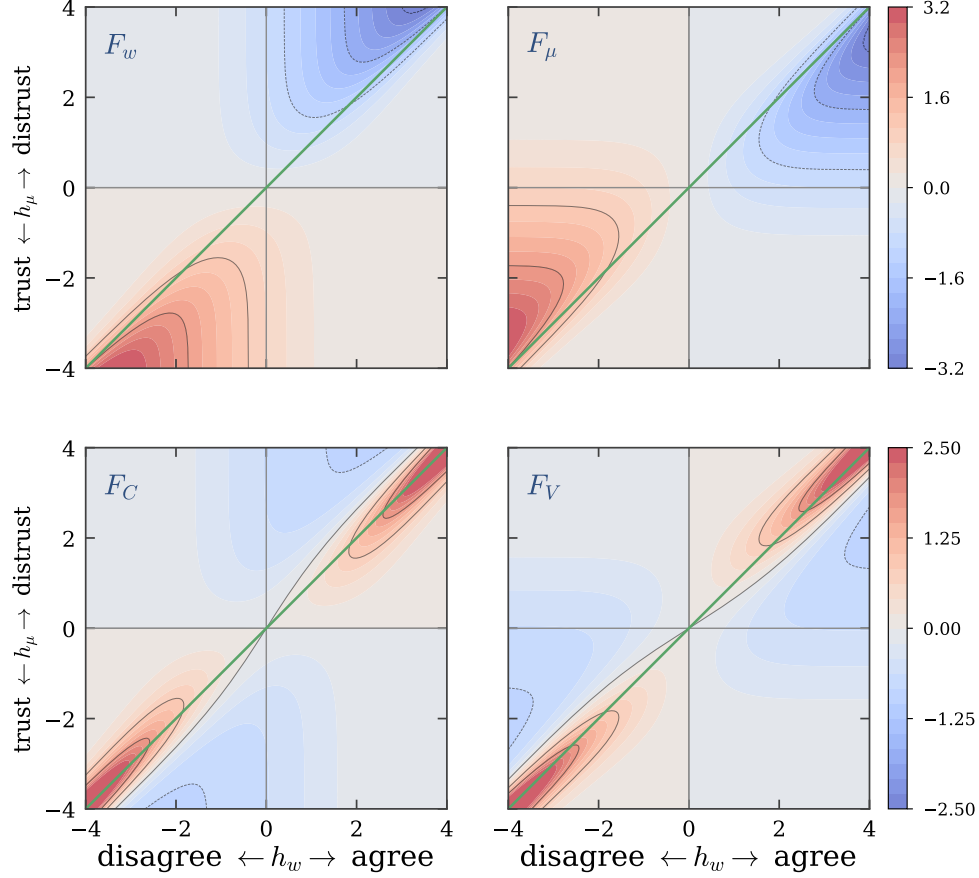


Figure 12: The two functions that move the means, F_w and F_μ , above (the pair shown in Figure 2) and the two that anneal the uncertainties, F_C and F_V , below. Each row shares a range. The mirror symmetry across $h_\mu = h_w$ holds in both rows: $F_w(x, y) = F_\mu(y, x)$ and $F_C(x, y) = F_V(y, x)$. F_C and F_V are negative almost everywhere, which is why both uncertainties shrink monotonically and the dynamics anneals (§3.1).

E MEASURING OPINION AND TRUST UPDATES UNDER IN-CONTEXT LEARNING

The two experiments use different tasks because they require different observables. The opinion experiment controls both fields and locates a decision threshold. The trust experiment elicits reliability before and after a message. Both use `gpt-5.6-luna` at low reasoning effort, with fixed model weights. The opinion experiment uses 10 323 calls and the trust experiment uses 3 916.

E.1 OPINION: F_w BY NULLING

Experimental design. We generated sixteen fictional archives and froze their descriptions before running the experiment. Each archive contains an institution, a type of record, an analyst, and pairs of invented entities. Using fictional material prevents the model from bringing factual knowledge to the task. After screening for a position bias, eleven archives remained.

Let A and B denote the two entities in a question. Three quantities change the prompt:

Quantity	Values	What changes in the prompt
s	$-4, -2, 0, 2, 4$	balance of the six initial records
k	$0, 4, 10, 16, 20$	analyst’s correct answers out of twenty
t	adaptive integers	number and direction of later records

The initial block always contains six records. Of these, $(6+s)/2$ support A and $(6-s)/2$ support B . Thus the five values of s give splits of 1:5, 2:4, 3:3, 4:2, and 5:1. Changing s changes the receiver’s initial opinion while holding the amount of evidence fixed. The value of k changes only the analyst’s stated record. Values below ten describe a usually wrong source, $k = 10$ is uninformative, and values above ten describe a usually correct source.

The quantity t is a measurement probe, not a third coordinate of the figure. When $t = 0$, the prompt contains no later evidence. When $t > 0$, it adds t new records that all support A ; when $t < 0$, it adds $|t|$ records that all support B . Each value of t produces a separate prompt. Starting at zero, we try progressively larger values in the direction needed to reverse the answer, then bisect the last interval. Sixteen binary answers are sampled at each value, and a probit fit

$$\Pr(A | t) = \Phi[\beta(t - t^*)] \quad (18)$$

places the decision threshold t^* between integer values. The inferred opinion before the later evidence is $\ell = -t^*$, measured in pieces of evidence. The model is never asked to report a probability or confidence.

For every archive, s , and k , the analyst either states that A is favored or that B is favored. Let $m = +1$ for the first direction and $m = -1$ for the second. We fit the threshold once with the analyst silent, giving ℓ_0 , and again with the statement present, giving ℓ_1 . These two prompts are otherwise identical. The entity order is also fixed within the pair and counterbalanced across conditions. The final design therefore contains $11 \times 5 \times 5 \times 2 = 550$ measured updates. Each update summarizes two threshold searches over several values of t ; it is not the response to one prompt.

Prompt example. The following is one prompt from the statement-present arm, with $s = 2$, $k = 16$, and $t = 3$. Thus, the prompt contains three later records supporting Durnel Lock. The structured response was restricted to Durnel Lock or Pavqet Lock. The text below is reproduced verbatim.

System message

You are an analyst working on a body of historical records. You judge each question only from the briefing you are given, and you always commit to the answer the briefing makes more likely, however slightly.

User message

The registry contains dated engineering entries for locks, sluices, and tow routes maintained by the Sornethic League. You are one of its analysts.

Question: which was opened earlier, Durnel Lock or Pavqet Lock?

You have examined 6 canal entries bearing on it:

Canal entry 1: indicates that Durnel Lock was opened earlier.
 Canal entry 2: indicates that Durnel Lock was opened earlier.
 Canal entry 3: indicates that Pavqet Lock was opened earlier.
 Canal entry 4: indicates that Pavqet Lock was opened earlier.
 Canal entry 5: indicates that Durnel Lock was opened earlier.
 Canal entry 6: indicates that Durnel Lock was opened earlier.

Registrar Veskul works on the same records. Of the last 20 questions of this kind whose answers were later established beyond doubt, Registrar Veskul answered correctly 16 times and incorrectly 4 times.

Registrar Veskul has now stated: "Durnel Lock was opened earliar."

A later consignment has since reached you: 3 further canal entries bearing on this question, of the same standing as any other record in the collection. All 3 indicate that Durnel Lock was opened earlier.

On the whole of the above, which was opened earlier, Durnel Lock or Pavqet Lock?

From thresholds to Figure 3. The labels s and k control where a point lies, but they are not plotted directly. Let H_λ denote the monotone conversion from evidence units to the probit fields used in (2). The scale λ is fixed from the analysts' accuracy records, not from the opinion updates. A separate nulling calibration confirms that four pieces of supplied evidence are recovered as 3.6–4.1 pieces, with a residual offset below one quarter of a piece. The testimony weights inferred from the five values of k correlate with their stated records at $r = 0.99$.

For a statement in direction m , the plotted horizontal coordinate and measured update are

$$h_w = mH_\lambda(\ell_0), \quad \Delta h_w = m[H_\lambda(\ell_1) - H_\lambda(\ell_0)]. \quad (19)$$

The factor m expresses both quantities relative to the statement: $h_w > 0$ means that the receiver initially agrees with it, and $\Delta h_w > 0$ means that the receiver moves toward it. Thus s controls the horizontal position, but the position is obtained from the silent threshold rather than assigned from s . Likewise, k controls the vertical position, but h_μ is obtained from the calibrated weight the model gives that track record rather than set equal to k . Variation across archives and threshold fits therefore leaves points with the same (s, k) near one another rather than at one exact coordinate.

The color of each point is Δh_w , the shift between the two fitted thresholds. It is not t : the sweep over t is only how we measure ℓ_0 and ℓ_1 . Nor do we identify Δh_w numerically with F_w . In (4), F_w is multiplied by the receiver's unobserved susceptibility along the issue direction. We therefore test $\Delta h_w \propto F_w(h_w, h_\mu)$ and fit one global positive scale α_w across all 550 updates. Figure 3 plots $\Delta h_w/\alpha_w$ against the raw analytic field. The evidence-to-probit conversion λ is calibrated independently and is not part of this fit. The reported sign agreement and rank correlation do not depend on α_w .

The experiment controls conviction only because the questions are fictional, and the results concern one model at one reasoning effort.

E.2 TRUST: F_μ BY ELICITED RELIABILITY

Stimulus construction and prompt controls. We use twenty open, low-valence themes. For each theme, the model first writes its own opinion and then generates and rates other statements from strong disagreement to strong agreement. We freeze these materials before the experiment.

Each prior context varies two quantities. First, the instruction says that the model holds its opinion either firmly or tentatively. Second, the model sees a history from another agent. We index this history by $k = n_+ - n_-$, where n_+ and n_- are the numbers of agreeing and disagreeing statements:

k	-5	-4	-3	-2	-1	0	1	2	3
n_+	0	0	0	0	1	1	2	2	3
n_-	5	4	3	2	2	1	1	0	0

Thus, negative k gives a mostly disagreeing history and positive k a mostly agreeing one. The prompt never states whether the other agent is reliable; the model must infer reliability from this

history. After the prior context, the other agent gives one new message. We use either a strongly agreeing message ($m = +1$) or a strongly disagreeing one ($m = -1$). History statements and new messages are drawn from separate pools.

How one context becomes two points. Before the new message, we ask for the probability r_0 that the other agent’s judgment is sound. We also ask for the probability c that a separate agent, described as sound, would express a view with which the model agrees. The latter is our direct proxy for the magnitude of the model’s conviction. After showing the new message, we ask again about the original agent and obtain r_m . We convert these three probabilities into the fields and the trust update as

$$h_\mu = -\Phi^{-1}(r_0), \quad h_w = m\Phi^{-1}(c), \quad \Delta h_\mu = -\Phi^{-1}(r_m) - h_\mu. \quad (20)$$

This gives the direct correspondence with Figure 3: h_w is the horizontal position, h_μ is the vertical position, and Δh_μ is the color. Each prior context therefore produces two points, one for each sign of the new message. They have the same vertical coordinate and opposite signs on the horizontal axis, but their colors can differ. Four of the 360 intended prior contexts lacked the required stimulus coverage, leaving 356 contexts and 712 plotted updates.

The plotted points measure Δh_μ , not F_μ itself. From (5), the two are proportional when the unobserved trust variance and susceptibility are held fixed. We therefore fit one global positive scale α_μ across all 712 updates and plot $\Delta h_\mu/\alpha_\mu$ against the raw analytic F_μ surface. The fitted scale does not affect the reported sign agreement or correlation.

Prompt example. The following after-message prompt is from the `difficulty` theme, with a balanced two-statement history and the firm conviction instruction. The before-message prompt is identical except that the paragraph beginning “They have now said” is absent. The model returned an integer from 0 to 100. The text is reproduced verbatim.

System message

You are an agent taking part in a discussion with other agents.

User message

Consider whether difficulty is a necessary part of a good education, or mostly an obstacle that good teaching should remove.

You currently have this opinion on that theme:

"Difficulty is a necessary part of good education when it is purposeful: grappling with challenging ideas builds understanding, resilience, and independence. However, good teaching should remove unnecessary obstacles so that students struggle with the subject itself, not with confusion, poor design, or lack of support."

You hold that opinion firmly.

You have discussed this theme with another agent 2 times before. On each occasion they said:

1. "Students learn best when information is presented clearly and directly, so struggle should be eliminated rather than treated as educationally valuable."
2. "Difficulty is valuable when it encourages students to think, question, and solve problems independently, but ineffective instruction should not be mistaken for rigor."

They have now said:

"Good education should preserve productive struggle while eliminating avoidable frustration; students learn most deeply when challenges are meaningful and support is available."

What is the chance, from 0 to 100, that this agent's judgement on this theme is sound?

Validation and limitations. As a check on the direct conviction proxy, we also ask for the probability q that the original agent’s next statement will agree with the model. Setting $e = 1 - r_0 = \Phi(h_\mu)$ in Equation (3) gives the indirect estimate

$$c_{\text{ind}} = \frac{q - e}{1 - 2e}. \quad (21)$$

This inversion is singular near neutral trust and can return values outside $[0, 1]$. It is admissible for only 280 of the 712 updates and correlates moderately with the direct proxy ($r = 0.46$). Figure 3 therefore uses the direct proxy, not this indirect estimate.

Three caveats matter. The direct question may mix conviction with perceived consensus. The histories vary from two to five statements as k changes, so history balance and length are not fully separated. Finally, all elicited probabilities are clipped to $[0.02, 0.98]$ before the probit transform, bounding individual field readings at about 2.05 in magnitude. The experiment therefore tests the quantitative relation between Δh_μ and F_μ over a restricted region, for one model and one reasoning effort.

F DERIVATION OF THE LEARNING ALGORITHM

The receiver’s belief before an interaction is the product of Gaussians (1). It observes σ_e , the emitter’s opinion on issue $\hat{x}$, and treats it as a report of an unknown true label σ_T transmitted through a channel that flips the label with probability $\varepsilon(z)$. Marginalizing the joint distribution of labels and of the issue as the emitter may have parsed it, y_t ,

$$L(\sigma_e | \hat{x}, u) = \sum_{\sigma_T} \int P(\sigma_e | \sigma_T, y_t, \hat{x}, u) P(\sigma_T | y_t, \hat{x}, u) P(y_t | \hat{x}, u) dy_t, \quad (22)$$

with $P(\sigma_e | \sigma_T, z) = (1 - \varepsilon(z))\delta_{\sigma_e, \sigma_T} + \varepsilon(z)\delta_{\sigma_e, -\sigma_T}$ and $P(\sigma_T | y_t, w) = \Theta(\sigma_T y_t \cdot w)$. Two modeling choices keep the algebra Gaussian. Taking $\varepsilon(z) = \Phi(z)$ with z Gaussian of mean $\mu_{e|r}$ and variance $V_{e|r}$ makes the flip probability a bounded function of a Gaussian variable while preserving conjugacy of the trust sector; a Beta distribution for ε , the tempting alternative, does not. Taking $P(y_t | \hat{x}, u)$ Gaussian with variance $v_p = \|w\|^{-2}$ lets an inexperienced agent, whose weights are small, be correspondingly unsure that it has parsed the issue as the emitter did. Together these give the likelihood

$$L(\sigma_e | z, \hat{x}, w) = \Phi(z) \Phi(-w \cdot \hat{x} \sigma_e) + (1 - \Phi(z)) \Phi(w \cdot \hat{x} \sigma_e), \quad (23)$$

and the covariance stays block diagonal in the two sectors if it starts so.

Averaging the likelihood over the current belief gives the evidence

$$Z(\sigma_e | \hat{x}, D_t) = \mathbb{E}_{Q_t}[L(\sigma_e | z, u, \hat{x})] = \sum_{\sigma_T} \mathbb{E}_z[P(\sigma_e | \sigma_T, \hat{x}, z)] \mathbb{E}_w[\Phi(\sigma_T w \cdot \hat{x})], \quad (24)$$

and the Gaussian integrals collapse onto the two scaled fields (2), giving (3): the evidence depends on the receiver’s state only through (h_w, h_μ) . This is the symmetry that makes the two sectors mirror images.

The exact Bayesian posterior is not Gaussian. Replacing it by the Gaussian of maximum entropy relative to the prior, subject to matching the first two moments of the posterior, gives updates in terms of derivatives of the log evidence,

$$\hat{w}_{t+1} = \hat{w}_t + C_t \frac{\partial \ln Z}{\partial \hat{w}}, \quad C_{t+1} = C_t + C_t \frac{\partial^2 \ln Z}{\partial \hat{w} \partial \hat{w}^\top} C_t, \quad (25)$$

and identically in the trust sector with $\mu_{e|r}, V_{e|r}$ in place of $\hat{w}, C$. Expressing the derivatives in the scaled fields yields (4)–(5) with the modulation functions (??). The first is a Hebbian term $\sigma_e \hat{x}$ with a tensorial, self-annealing rate; the rest is the online estimation of the channel.

G GENEALOGY OF THE LEARNING RULE

The update (4)–(5) descends from the variational optimization of modulated Hebbian on-line learning (Kinouchi & Caticha, 1992; Kinzel & Ruján, 1990), given a Bayesian reading by Oppen (1996; 1998) and applied across architectures by Solla & Winther (1999); the entropic-dynamics formulation we use is Caticha (2020). Learning through a noisy channel whose noise level must itself be inferred was analyzed by Kinouchi & Caticha (1993) and Biehl et al. (1995); Alves & Caticha (2016) identified that inferred level with distrust and made it dynamical, with applications to voting

(Caticha et al., 2015) and to panels of appellate judges (Caticha & Alves, 2019), and a mean-field treatment by Simões & Caticha (2018). The blame-attribution behavior resembles the least-action rule of Mitchison & Durbin (1989), which suffers retarded learning (Kabashima, 1994), whereas the optimized algorithm attains generalization error decaying as α^{-1} even under output noise (Simonetti & Caticha, 1996).

H THE GROUP-BIAS FIELD

Equation (8) is one filling of a 2×2 matrix D indexed by the classes of receiver and emitter, $h_w^p = h_w + D_{\kappa_r, \kappa_e}$. Four numbers are free. Since κ_r and κ_e each take two values, any filling decomposes uniquely into four orthogonal components,

$$D_{\kappa_r, \kappa_e} = a + b\kappa_r + c\kappa_e + p\kappa_r\kappa_e, \quad (26)$$

with independent magnitudes and one meaning each. a shifts perceived agreement by the same amount whoever is speaking: a uniform credulity, referring to no label. b depends only on who is *listening*, so one class is systematically the more credulous. c depends only on who is *speaking*, so one class is believed more by everyone, its own members included. p depends only on whether the two classes *match*, which is in-group favoritism and out-group hostility at once, and is what (8) uses with $a = b = c = 0$.

Relabeling the classes sends $(a, b, c, p) \rightarrow (a, -b, -c, p)$. Only a and p survive it, and a does not refer to the label, so p is the one component that is both class-dependent and symmetric between the classes. That is why the main text uses it: under p alone the two classes are interchangeable, so any asymmetry in the outcome is spontaneous rather than imposed. The components that change sign, b and c , name a class rather than a relation between classes.

None of the four is redundant. The components are distinct because the modulation functions have a fixed scale. In particular, the uniform shift a moves the trust separatrix but not the opinion separatrix.

The four components pair with four ways trust can carry a label. Because trust is directed, its class dependence can be decomposed into the same four channels: uniform, receiver-specific, emitter-specific, and relational.

b and c are transposes, and nothing here can tell them apart. Receiver- and emitter-specific fields transpose the trust matrix. The symmetrized order parameters used in the main text cannot distinguish them.

What we have not run. The main text studies only p , with $a = b = c = 0$. We have not simulated the other components and make no claim about their collective phases.

I THE SIGN OF THE GROUP-BIAS FIELD

The sign of the group-bias field is fixed by the algorithm and not free. By (??), $F_\mu = (1 - 2\Phi(h_w))g(h_\mu)/Z$ is negative when $h_w > 0$, so perceived agreement drives distrust μ down. A receiver that adds $+p$ to h_w therefore becomes more trusting (tolerant) and one that adds $-p$ becomes more distrustful (intolerant). Equation (8) with $p > 0$ consequently gives in-group trust and out-group distrust, i.e. the discriminatory phase, which is the convention used throughout.

The companion manuscript (Anonymous, 2026) tabulates the in-group entry as $-p$ while its text and figures place the discriminatory phases at $p > 0$; those two statements are inconsistent, and taken literally the tabulated signs put the discriminatory phases at $p < 0$, mirroring every map in p . The same inconsistency appears in its description of the learning flow, where the panel labeled $D_{e|r} = -p$ is called the more tolerant one although $-p$ is the intolerant entry, and where the “distrust and disagree” basin is said to grow for $p > 0$ although adding $+p$ shrinks it. A single global sign reconciles everything; we adopt it in the form that leaves (8) untouched. The accompanying code implements both readings behind a flag, and Figure 13 runs the same sweep under each: the two rows are mirror images in p , and only the first matches the maps as published.

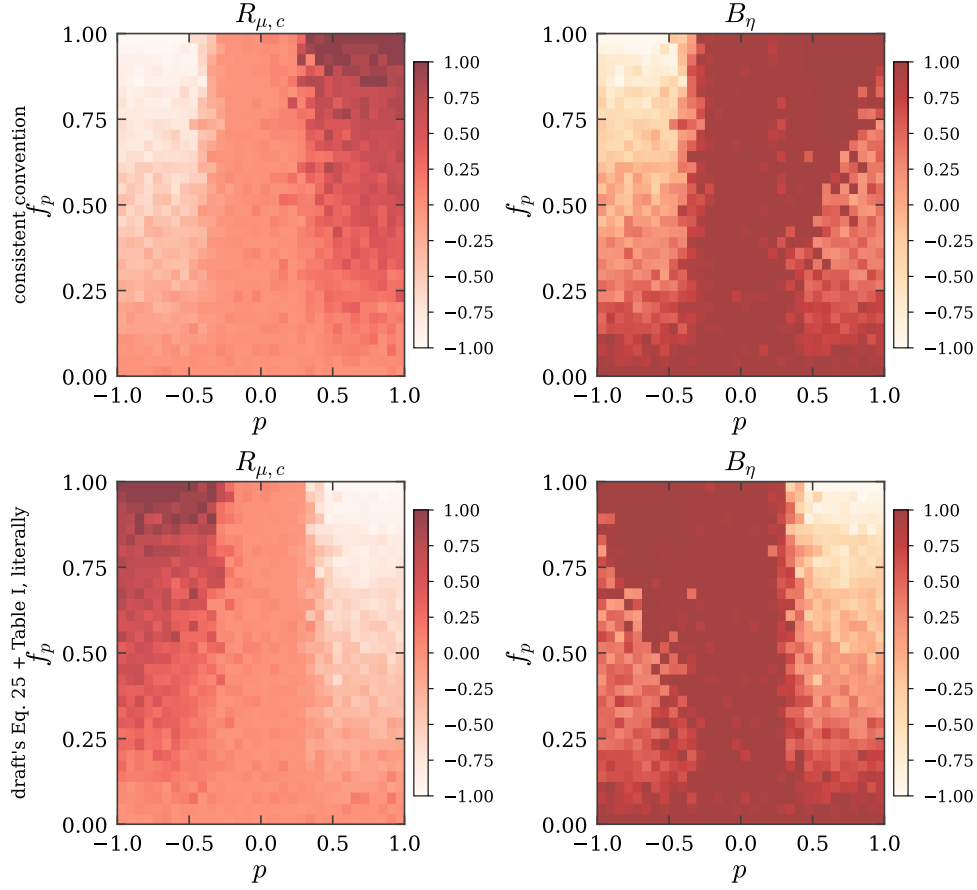


Figure 13: **The two readings of the group-bias field.** Trust-class correlation and trust balance over the (p, f_p) plane under the convention used in this paper (top) and under the companion manuscript’s table read literally with its Eq. 25 (bottom). The physics is identical; which half of the p axis is discriminatory is not.

J CONVENTIONS IN THE ORDER PARAMETERS

Two choices in (12) deserve statement.

The class indicator. We use $G_{IJ} = \kappa_I \kappa_J \in \{-1, +1\}$. A $\{0, 1\}$ indicator, which counts only same-class pairs, cannot represent anti-correlation: under perfect reverse discrimination it returns $-(N/2 - 1)/(N - 1)$ rather than -1 , and its range depends on N . Since the reverse-discrimination half of the phase diagram is a substantive part of the result, the signed indicator is the appropriate one. Both are implemented.

Normalization. All three averages in (10) and (12) run over the $N(N - 1)/2$ unordered pairs, and the $\frac{1}{2}$ on the symmetrized trust puts all three in $[-1, 1]$. This matters because the three are read together as the color channels of Figure 1; a literal reading of the model as previously written normalizes $R_{c,w}$ by $N(N - 1)$ instead, capping it at $1/2$ and making the channels incommensurate. Both are implemented.

K PARAMETERS

The model needs six numbers. One is recovered from the companion manuscript, one is calibrated against its published figures, two are prior conventions, and two are free choices; all six are listed with their provenance rather than left implicit, since the manuscript states none of them. The accompanying code exposes each of them and its calibration script reproduces the comparisons below.

symbol	meaning	value	provenance
K	dimension of the issue embedding	30	recovered (see below)
N	agents in the society	40	chosen; converged (see below)
P	issues on the agenda	5 / 100	chosen: $\alpha = P/K$ either side of 1
Δt	interactions per ordered pair	500	calibrated (see below)
C_0, V_0	initial uncertainties	$I, 1$	uninformative prior
μ_0	initial distrust	$\mathcal{U}(-1, 1)$	gives $B_\rho \approx B_\eta \approx 0$ at $t = 0$

K is recovered, not chosen. The companion manuscript reports balance trajectories at $\alpha \in \{0.03, 0.17, 0.23, 0.33, 0.50, 0.67, 1.67, 3.33, 333.33\}$. These are exactly $P/30$ for $P \in \{1, 5, 7, 10, 15, 20, 50, 100, 10^4\}$, which fixes $K = 30$.

Δt is calibrated against the published trajectories. The dynamics anneals rather than reaching a stationary state, so the measurement time matters and the companion manuscript does not state it (its text carries a literal placeholder in place of the value). We scan Δt and compare the endpoints of all nine balance trajectories against the published ones:

Δt (interactions per ordered pair)	60	125	250	500	1000
rms distance to published endpoints	0.63	0.31	0.15	0.13	0.15

The minimum is flat-bottomed and sits at $\Delta t = 500$, which we use throughout. Seven of the nine endpoints then agree to within about 0.1, and B_ρ matches the three largest agendas almost exactly, but no single Δt reproduces all nine: two middle curves ($\alpha = 0.23, 0.33$) sit above ours, and the two largest agendas have a lower published B_η than we obtain at any Δt that fits the rest. Either the society size differs from ours or those middle curves, which overlap within a few pixels in the printed figure, cannot be digitized reliably; we cannot distinguish the two. Dropping the ambiguous readings does not move the selected value. All the Δt -independent checks pass: the above/below-diagonal ordering with its crossover at $\alpha \approx 1.7$, the sharp sign flip of $R_{\mu,c}$ at $p \approx 0$, and the $R_{c,w}$ wedge confined to large p and large f_p .

$N = 40$ is large enough. Repeating five representative points of the plane at $N \in \{20, 30, 40, 60\}$ with twelve realizations each, the order parameters at $N = 60$ differ from those at $N = 40$ by at most 0.02 ($R_{\mu,c}$, $R_{c,w}$), 0.03 (B_ρ) and 0.05 (B_η), while $N = 20$ differs by up to 0.24. The maps are therefore converged in N at $N = 40$, and the cost of going further is steep: total work grows as N^3 .

Numerical safeguards. Three floors are applied, none of which the model prescribes: on the evidence Z , whose exact modulation functions diverge in the deep dissonance corners; on the trust variance V , which $F_V < 0$ drives monotonically toward zero; and on the covariance update, clipped to the boundary at which $C + a(C\hat{x})(C\hat{x})^\top$ ceases to be positive semi-definite. The last is the only one that can alter a trajectory, and it never triggered in the runs reported here.